Network Services

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Chapter 1:
IPv4 Addressing and Subnetting

Learning Objectives

At the end of this chapter, students will be able to:

- Recall the network fundamentals: OSI and TCP/IP models, encapsulation, addressing, devices, and protocols.
- Explain the structure of IP addressing and convert between 8-bit binary and decimal numbers.
- Given an IPv4 address, classify it by type and describe how it is used in the network.
- Determine the network portion of a host address and explain the role of the subnet mask.
- Given IPv4 addressing information and design criteria, calculate the appropriate addressing components (subnetting).
- Use common utilities (ping, traceroute) to verify connectivity and the IP protocol stack of a host.
- Compare classful and classless addressing, and explain VLSM and CIDR.

Recap - What Is a Network?

- A **network** is a set of interconnected devices that **communicate** and **share resources** (data, printers, Internet access...).
- Every network is built from **three components**: end devices, intermediary devices, and network media.
- Networks are classified by their scope:
  - **LAN** (Local Area Network): a building, a campus; owned by one organization.
  - **WAN** (Wide Area Network): connects LANs over large distances; often via an ISP.
  - **Internet**: the worldwide network of interconnected networks.

End devices
PCs, laptops, servers, phones, printers

Intermediary devices
Switches, routers, access points, firewalls

Network media
Copper (UTP), fiber optic, wireless

Recap - The OSI Model

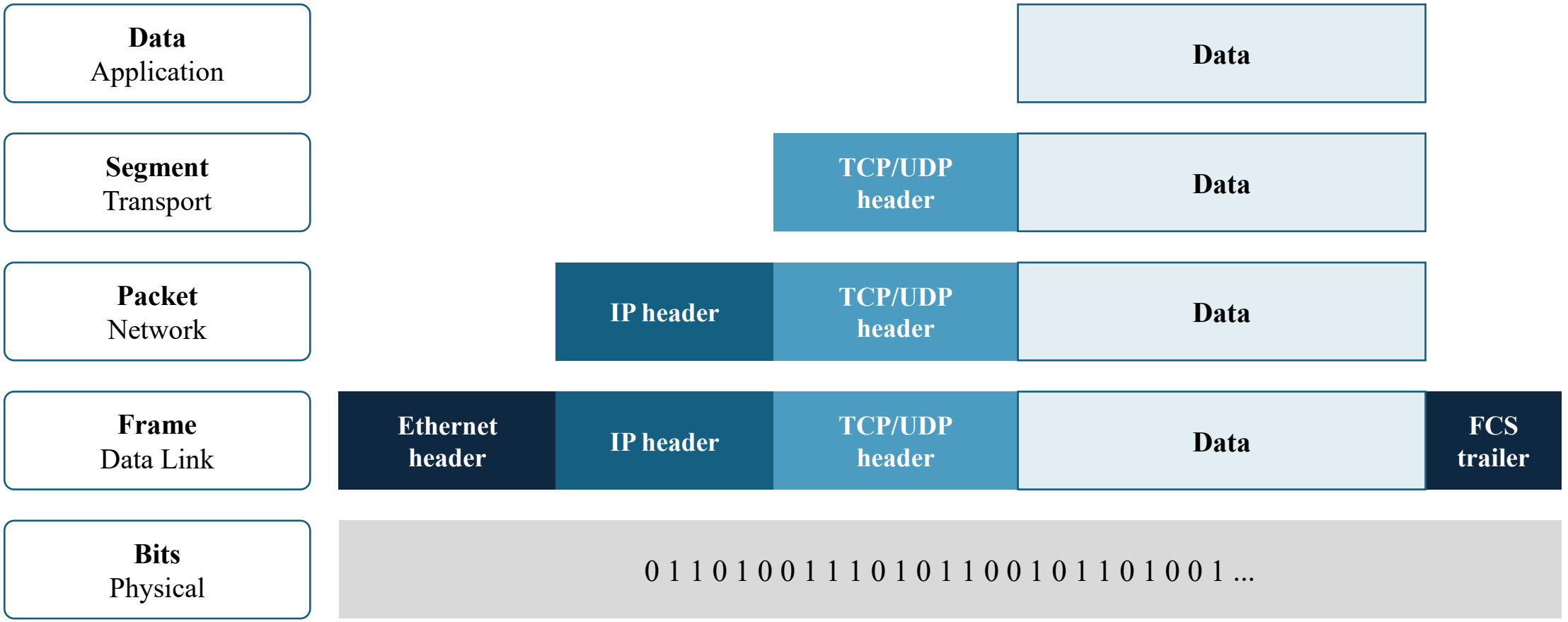
#	Layer	Role	PDU	Examples
7	Application	Interface with user applications	Data	HTTP, DNS, DHCP, FTP
6	Presentation	Format, encryption, compression	Data	TLS, JPEG, ASCII
5	Session	Open, manage and close sessions	Data	NetBIOS, RPC
4	Transport	Process-to-process delivery, ports, reliability	Segment	TCP, UDP
3	Network	Logical addressing and routing	Packet	IPv4, IPv6, ICMP
2	Data Link	Physical (MAC) addressing, access to the medium	Frame	Ethernet, Wi-Fi
1	Physical	Transmission of bits on the medium	Bits	Cables, signals, hubs

Layer 3 (Network) is where IPv4 addressing works: this is the focus of today's chapter.

Recap - OSI Model vs TCP/IP Model

OSI Model	TCP/IP Model	Main protocols
7. Application	Application	HTTP, HTTPS, DNS, DHCP, FTP, SSH, SMTP
6. Presentation		
5. Session		
4. Transport	Transport	TCP, UDP
3. Network	Internet	IPv4, IPv6, ICMP
2. Data Link	Network Access	Ethernet, Wi-Fi (802.11), cables
1. Physical		

Recap - Encapsulation



Each layer adds its header on the way down (**encapsulation**) and removes it on the way up (**decapsulation**). The **IP header** carries the source and destination IPv4 addresses.

Recap - Three Types of Addresses

Address	OSI layer	Size	Example	Identifies
MAC address	2 - Data Link	48 bits (hexadecimal)	00:1A:2B:3C:4D:5E	A network interface in the local network
IP address (IPv4)	3 - Network	32 bits (dotted decimal)	192.168.1.10	A host and its network, end to end
Port number	4 - Transport	16 bits (0 - 65535)	80, 443, 53	An application / process on the host

Remember: the **MAC** addresses change at every router (hop by hop), while the **source and destination IP** addresses stay the same from end to end.

Recap - Network Devices

Device	OSI layer	Forwards according to	Collision domain	Broadcast domain
Hub	1 - Physical	Nothing: repeats bits on all ports	One for all ports	One
Switch	2 - Data Link	MAC address table	One per port	One (by default)
Router	3 - Network	IP address / routing table	One per interface	One per interface

Key idea: only a **router** (Layer 3) separates **broadcast domains**. This is the basis of **subnetting** (Section 3).

Recap - TCP vs UDP

Criterion	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Acknowledgements and retransmission	No guarantee (best effort)
Ordering	Segments are re-ordered	No ordering
Overhead / speed	Larger header (20 bytes), slower	Small header (8 bytes), faster
Typical uses	Web (HTTP/S), e-mail, file transfer, SSH	DNS, DHCP, VoIP, video streaming, games

IP is **best effort**: when reliability is needed, it is provided by **TCP** at the transport layer.

Recap - Common Protocols and Ports

Protocol	Port	Transport	Role
HTTP / HTTPS	80 / 443	TCP	Web (HTTPS = encrypted with TLS)
DNS	53	UDP (and TCP)	Translates domain names into IP addresses
DHCP	67 / 68	UDP	Automatically assigns IP address, mask, gateway, DNS
FTP	20 / 21	TCP	File transfer
SSH / Telnet	22 / 23	TCP	Remote access (SSH is encrypted, Telnet is not)
SMTP	25	TCP	Sending e-mail
POP3 / IMAP	110 / 143	TCP	Receiving e-mail

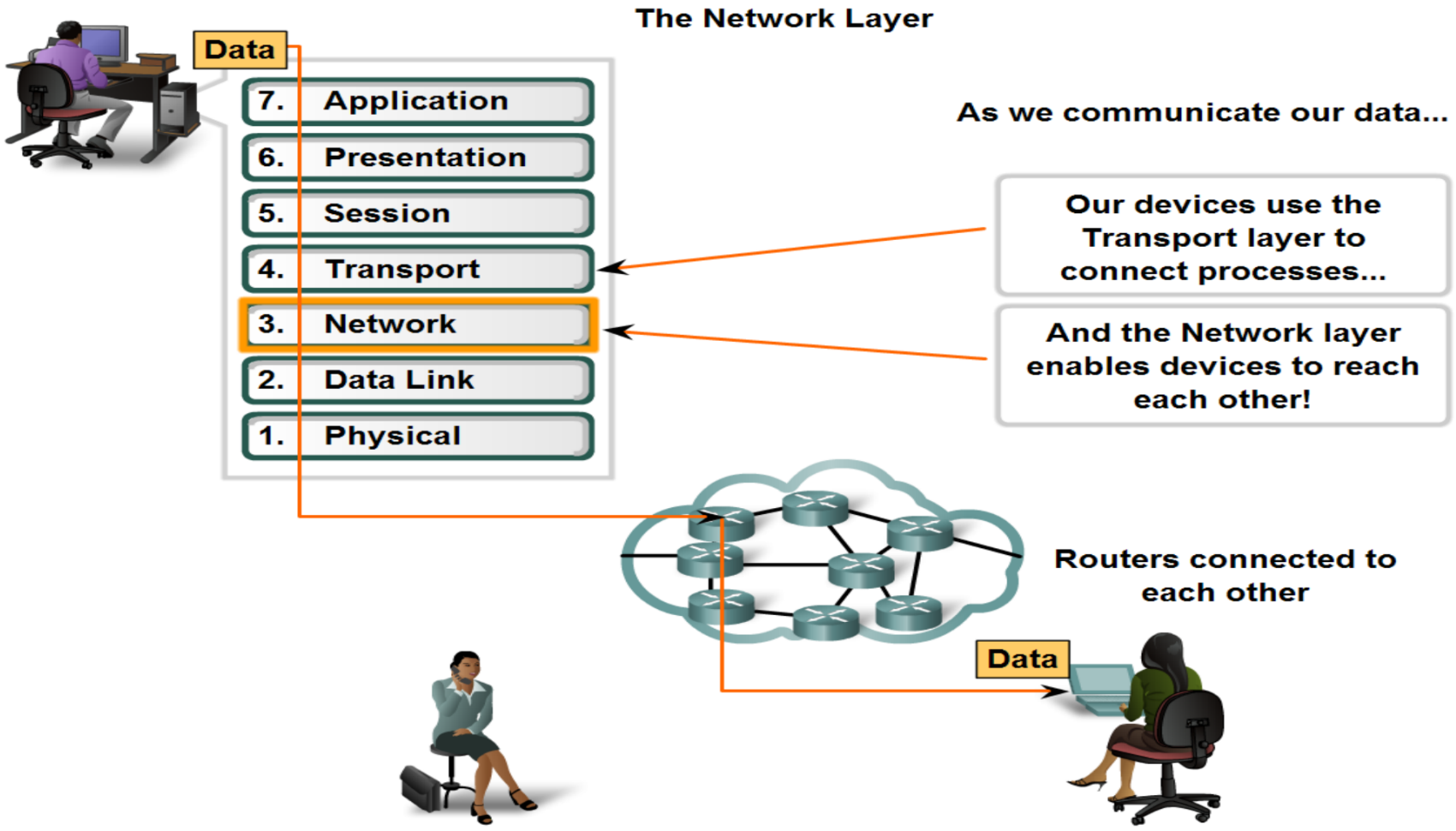
DHCP provides the 4 parameters we study in this chapter: **IP address, subnet mask, default gateway, DNS server**.

Recap - Warm-up Quiz

Answer these questions:

1. At which OSI layer does a router work?
2. What is the name of the PDU at Layer 2?
3. When a packet crosses several routers, which addresses change at each hop: MAC or IP?
4. Which transport protocol and port does DNS mostly use?
5. Does a switch separate broadcast domains? Why?
6. Name the four layers of the TCP/IP model.
7. Which header contains the source and destination IP addresses, and at which layer is it added?

1. IPv4 Address

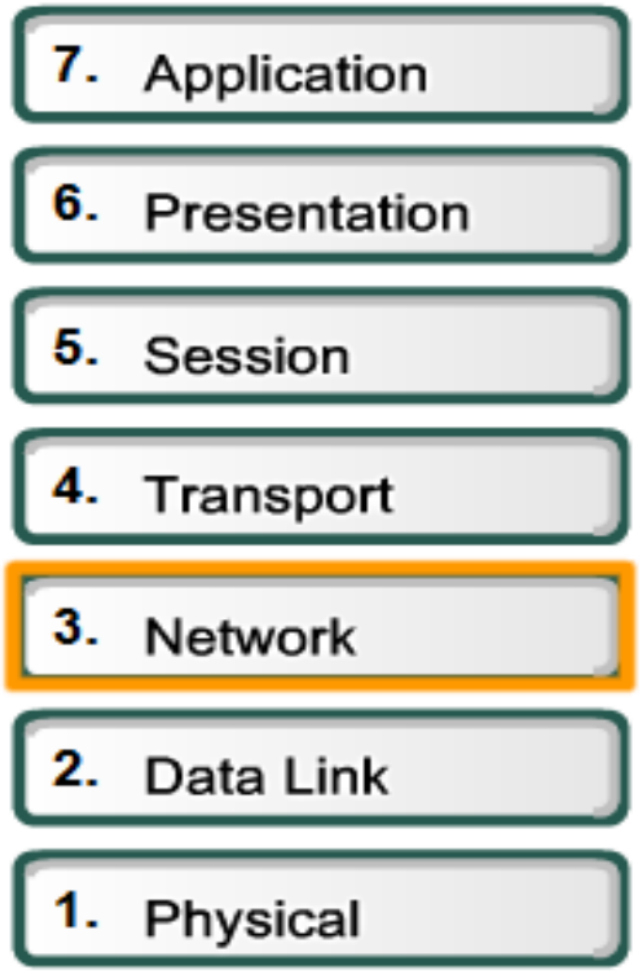


1. IPv4 Address

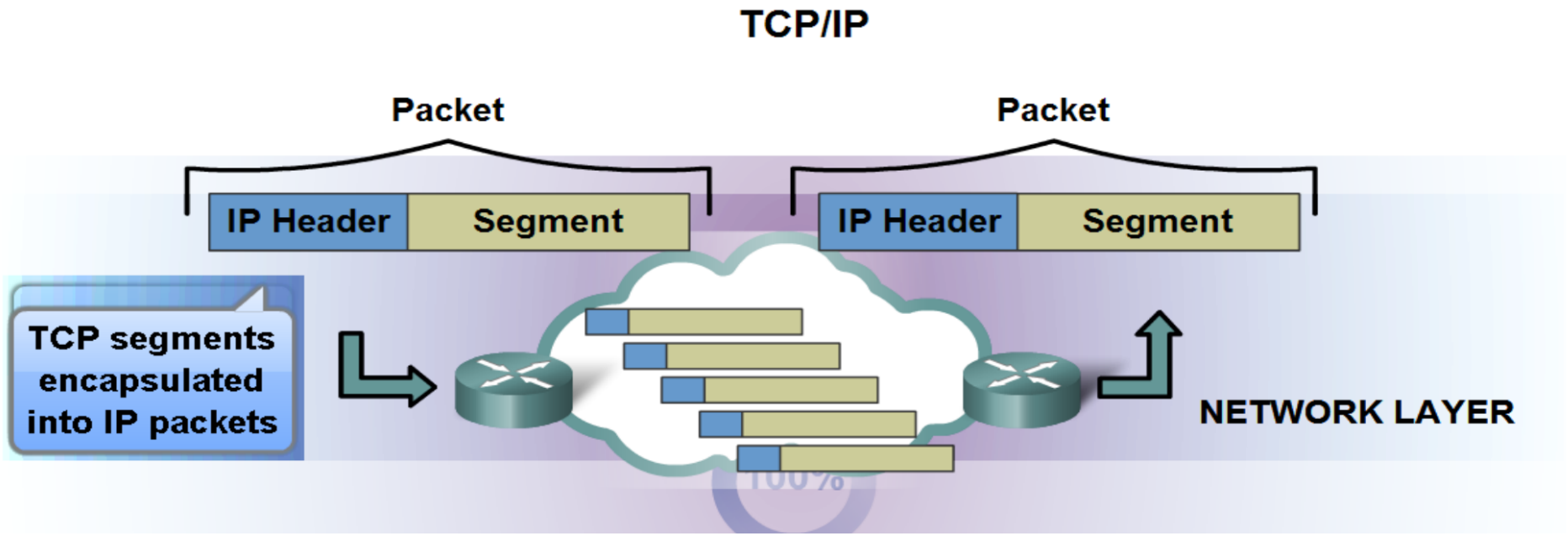
Network Layer Protocols

- Internet Protocol version 4 (IPv4)
- Internet Protocol version 6 (IPv6)
- Novell Internetwork Packet Exchange (IPX)
- AppleTalk
- Connectionless Network Service (CLNS/DECnet)

Key idea: IPv4 and IPv6 are the network layer protocols used today; this chapter focuses on IPv4.

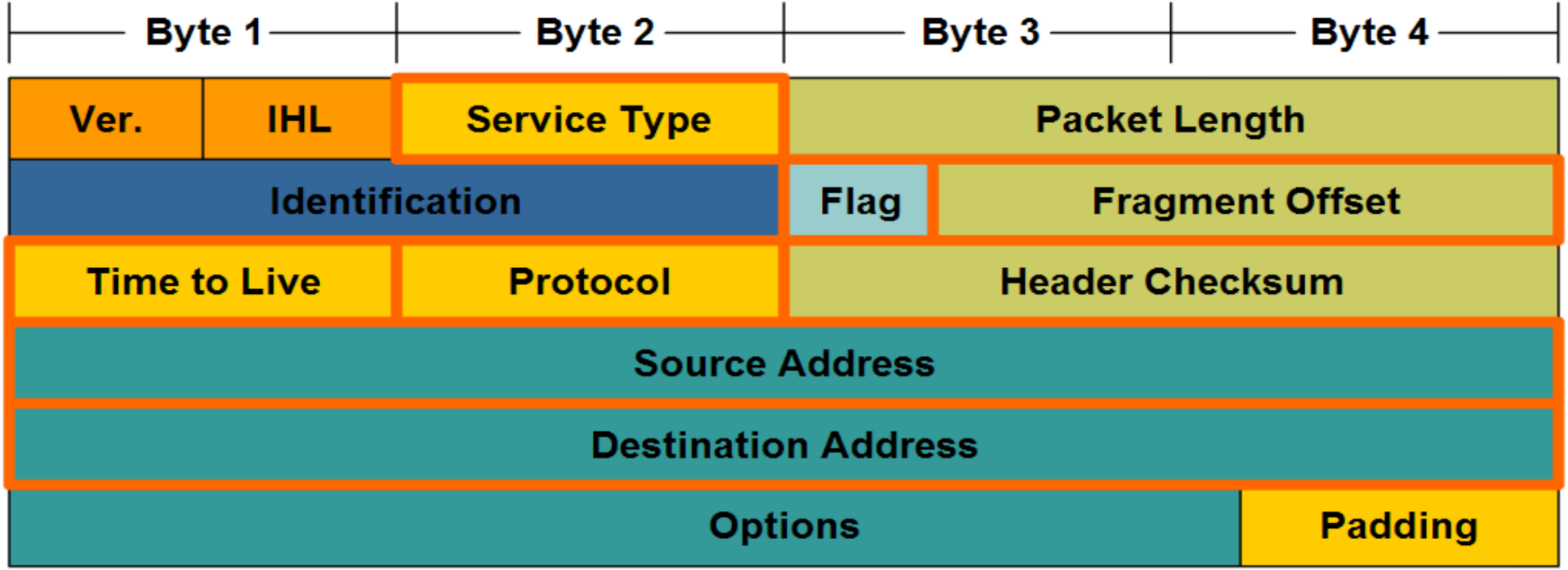


1. IPv4 Address - Internet Protocol (IP)



- **Connectionless:** no connection is established before sending data packets.
- **Best Effort (unreliable):** no overhead is used to guarantee packet delivery.
- **Media Independent:** operates independently of the medium carrying the data.

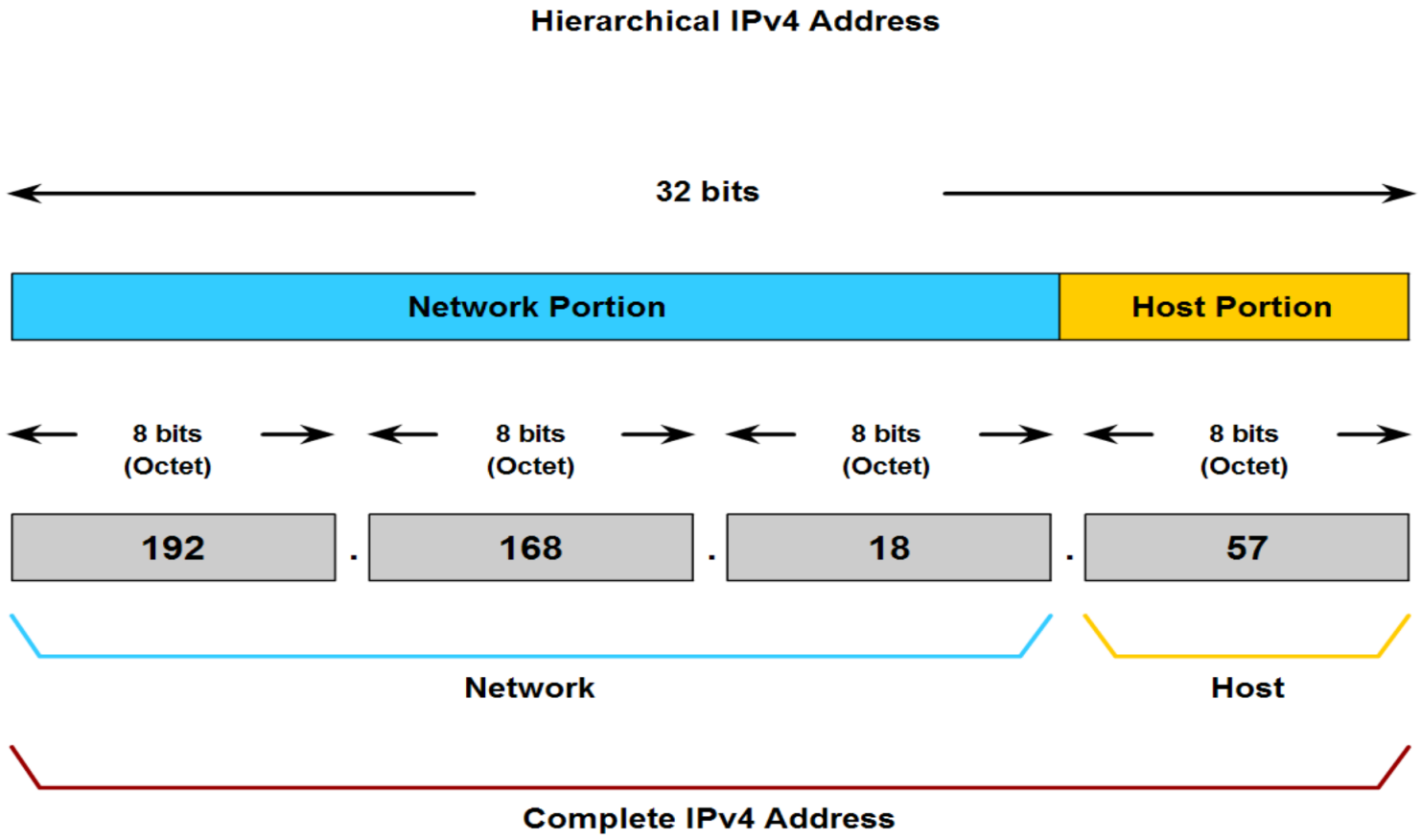
1. IPv4 Address - IPv4 Packet Header



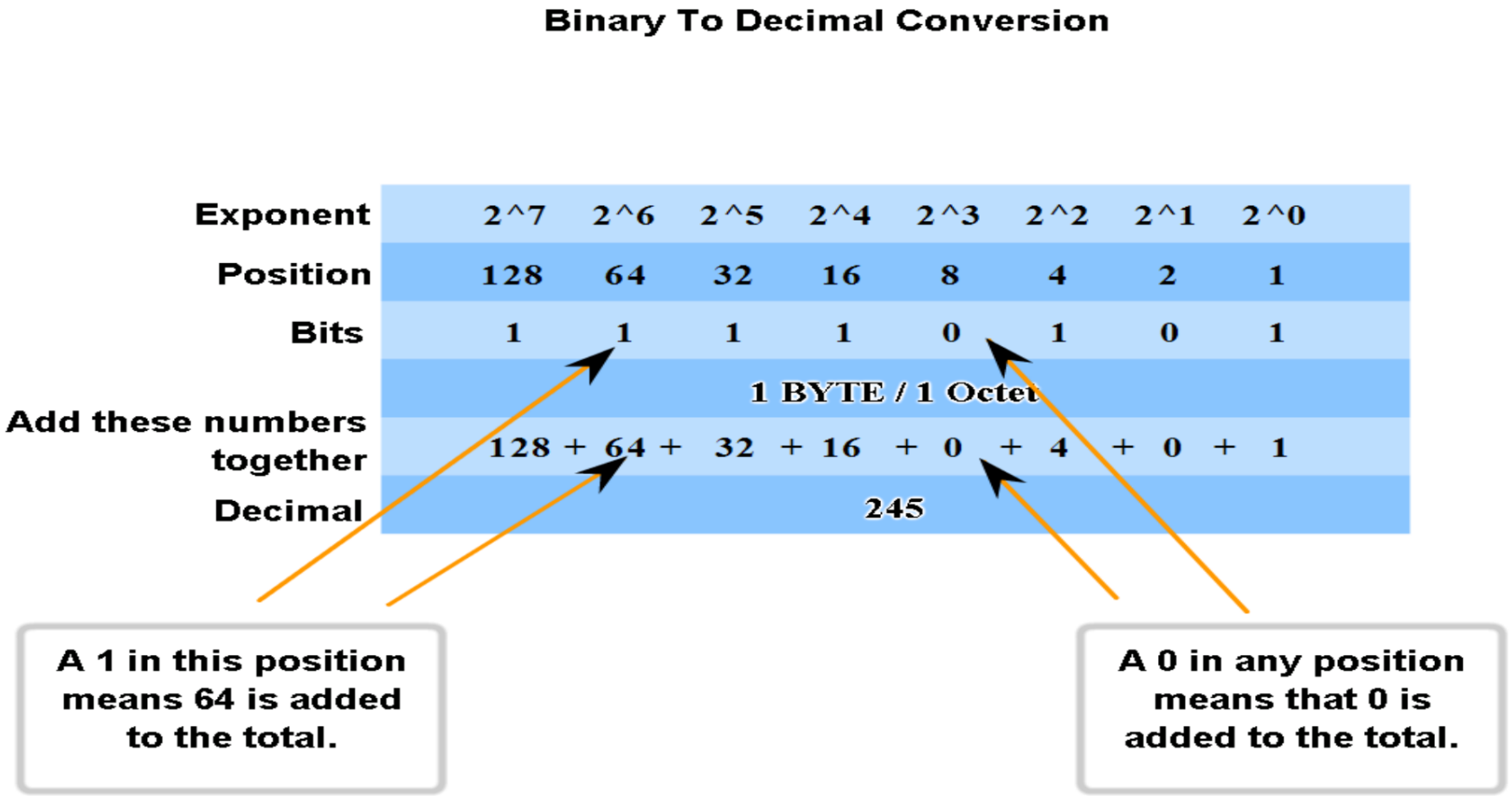
Key fields: **Source/Destination Address** (32 bits each), **Time to Live** (hop limit), **Protocol** (TCP, UDP, ...), **Flag / Fragment Offset** (fragmentation).

1. IPv4 Address - Hierarchical Addressing

An IPv4 address is **32 bits** = 4 octets, written in dotted decimal. It has two parts: the **network portion** and the **host portion**.



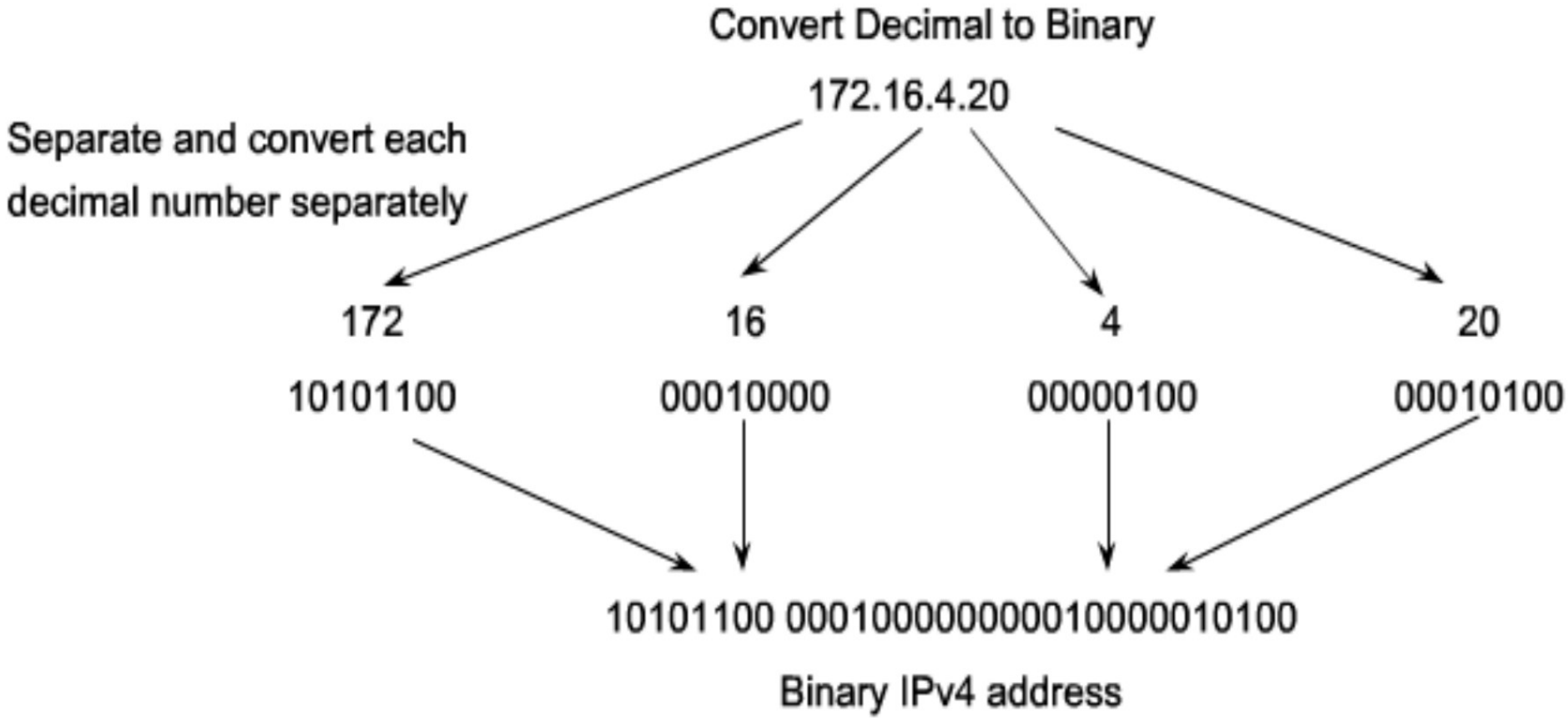
1. IPv4 Address - Binary & Decimal



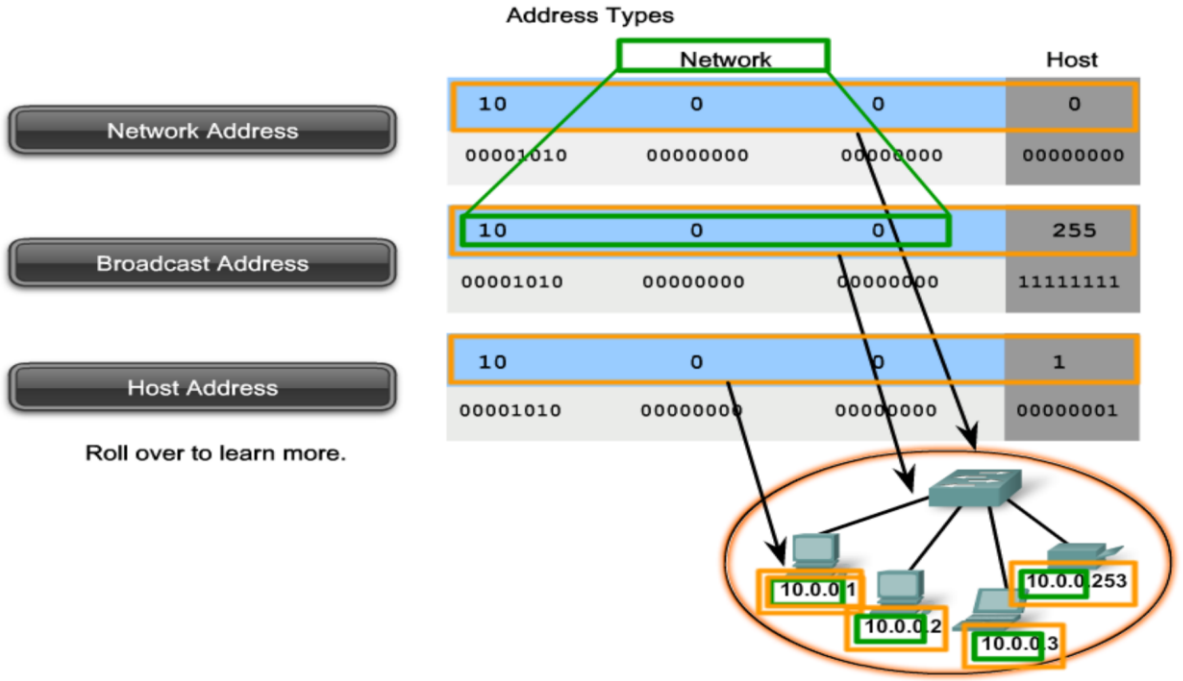
11110101 in binary = 128 + 64 + 32 + 16 + 4 + 1 = **245** in decimal

1. IPv4 Address - Binary & Decimal

Separate the address into four decimal numbers and convert **each one separately** into 8 bits, then concatenate them.



2. Addressing the Network - Address Types



- **Network address:** all host bits = 0 (e.g., 10.0.0.0).
- **Broadcast address:** all host bits = 1 (e.g., 10.0.0.255).
- **Host addresses:** everything in between (10.0.0.1 → 10.0.0.254).

2. Addressing the Network - IP Address Classes

Class	1st octet range	1st octet bits	Network (N) / Host (H)	Default mask	Networks / hosts per network
A	1 - 127*	0xxxxxxx	N.H.H.H	255.0.0.0 (/8)	128 nets (2^7) 16,777,214 hosts (2^24 - 2)
B	128 - 191	10xxxxxx	N.N.H.H	255.255.0.0 (/16)	16,384 nets (2^14) 65,534 hosts (2^16 - 2)
C	192 - 223	110xxxxx	N.N.N.H	255.255.255.0 (/24)	2,097,152 nets (2^21) 254 hosts (2^8 - 2)
D	224 - 239	1110xxxx	NA (multicast)	-	-
E	240 - 255	1111xxxx	NA (experimental)	-	-

* 127.x.x.x is reserved for loopback. In the host part, all zeros (0) and all ones (1) are not valid host addresses.

2. Addressing the Network - Subnet Mask

- The subnet mask identifies the **network portion** (bits = 1) and the **host portion** (bits = 0).
- **Network address = IP address AND subnet mask.**
- The prefix (/16) is the number of bits set to 1 in the mask.

Applying the Subnet Mask

A device with address 192.0.0.1 belongs to network 192.0.0.0

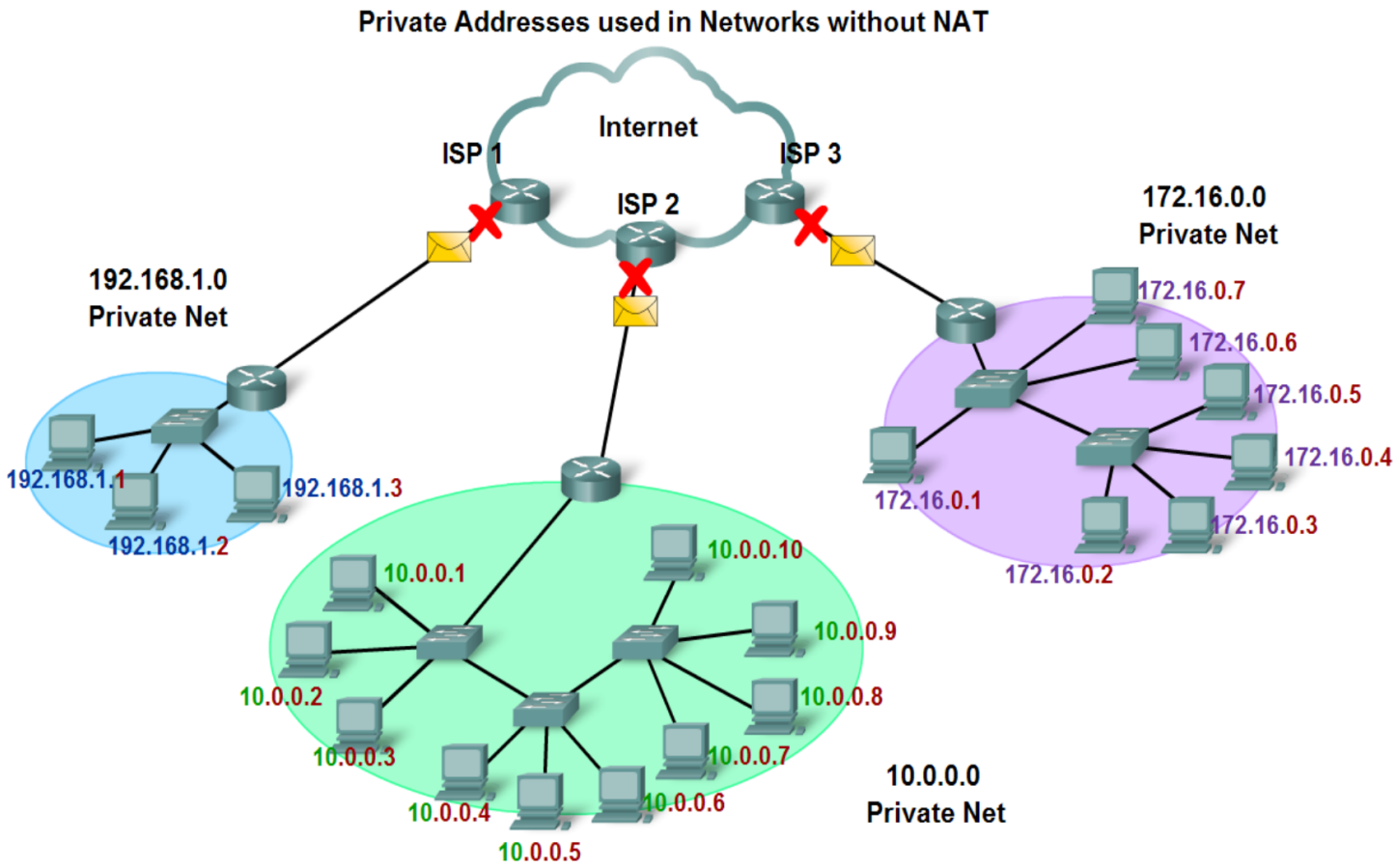
	High order bits Prefix /16		Low order bits	
	192 .	0 .	0 .	1
Host Address	11000000	00000000	00000000	00000001
Subnet Mask	255 11111111	255 11111111	0 00000000	0 00000000
Network Address	11000000	00000000	00000000	00000000
Network	192 .	0 .	0 .	0

2. Addressing the Network - Private IP Addresses

- Private IP addresses are another solution to the problem of the exhaustion of public IP addresses.
- Public networks require hosts to have **unique** IP addresses; private networks may use any address of these ranges, as long as each host is unique **within** the private network.
- Private addresses are for intra-network use only and **cannot be routed across the Internet**.

Class	RFC 1918 internal address range	Prefix
A	10.0.0.0 to 10.255.255.255	10.0.0.0/8
B	172.16.0.0 to 172.31.255.255	172.16.0.0/12
C	192.168.0.0 to 192.168.255.255	192.168.0.0/16

2. Addressing the Network - Private Nets without NAT



Packets with private addresses are **dropped by the ISPs**: to reach the Internet, a private network needs **NAT**.

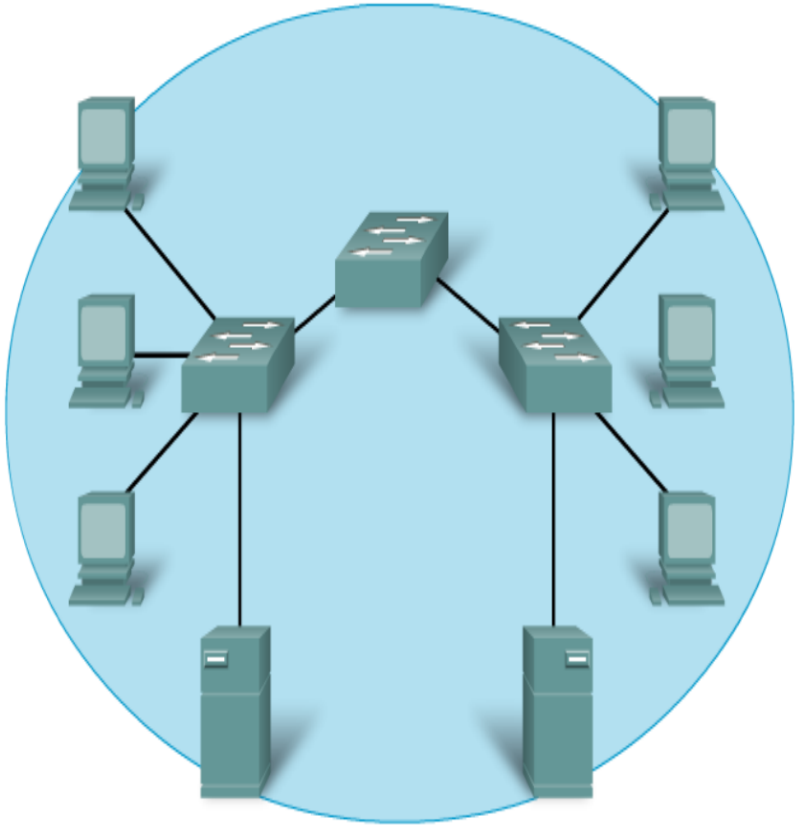
3. IPv4 Subnetting - Segmentation of Networks

The purposes of network segmentation are to:

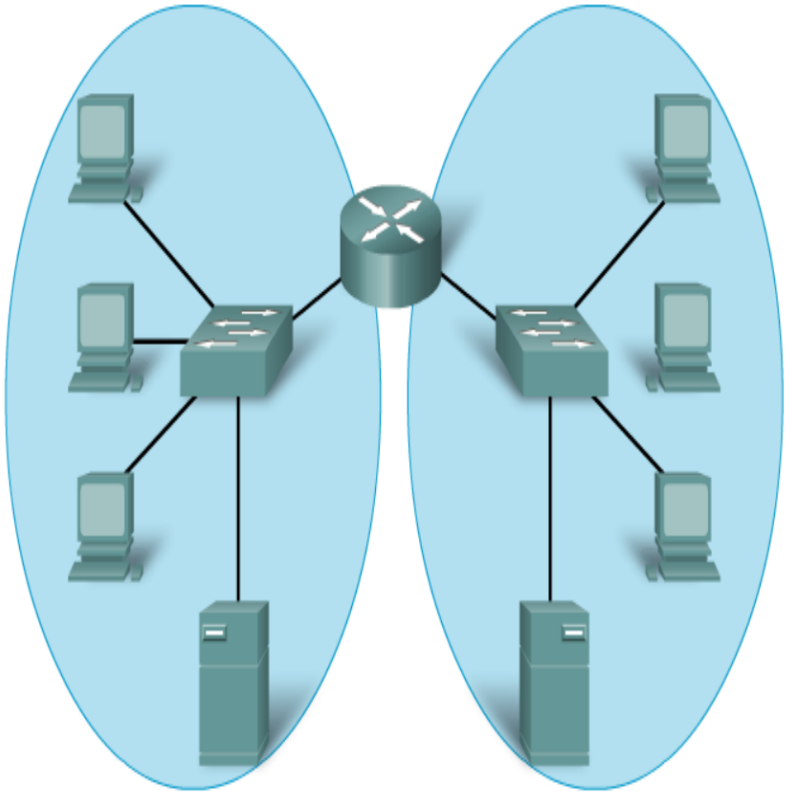
- **Increase network performance**: smaller broadcast domains.
- **Increase network security**: isolate groups of users and control traffic between them.
- **Increase network manageability**: smaller, organised networks are easier to administer.

Key idea: because of the fast deployment of the Internet, IP addresses became insufficient. Subnetting divides one network into several smaller networks and avoids wasting addresses.

3. IPv4 Subnetting - Increase Network Performance

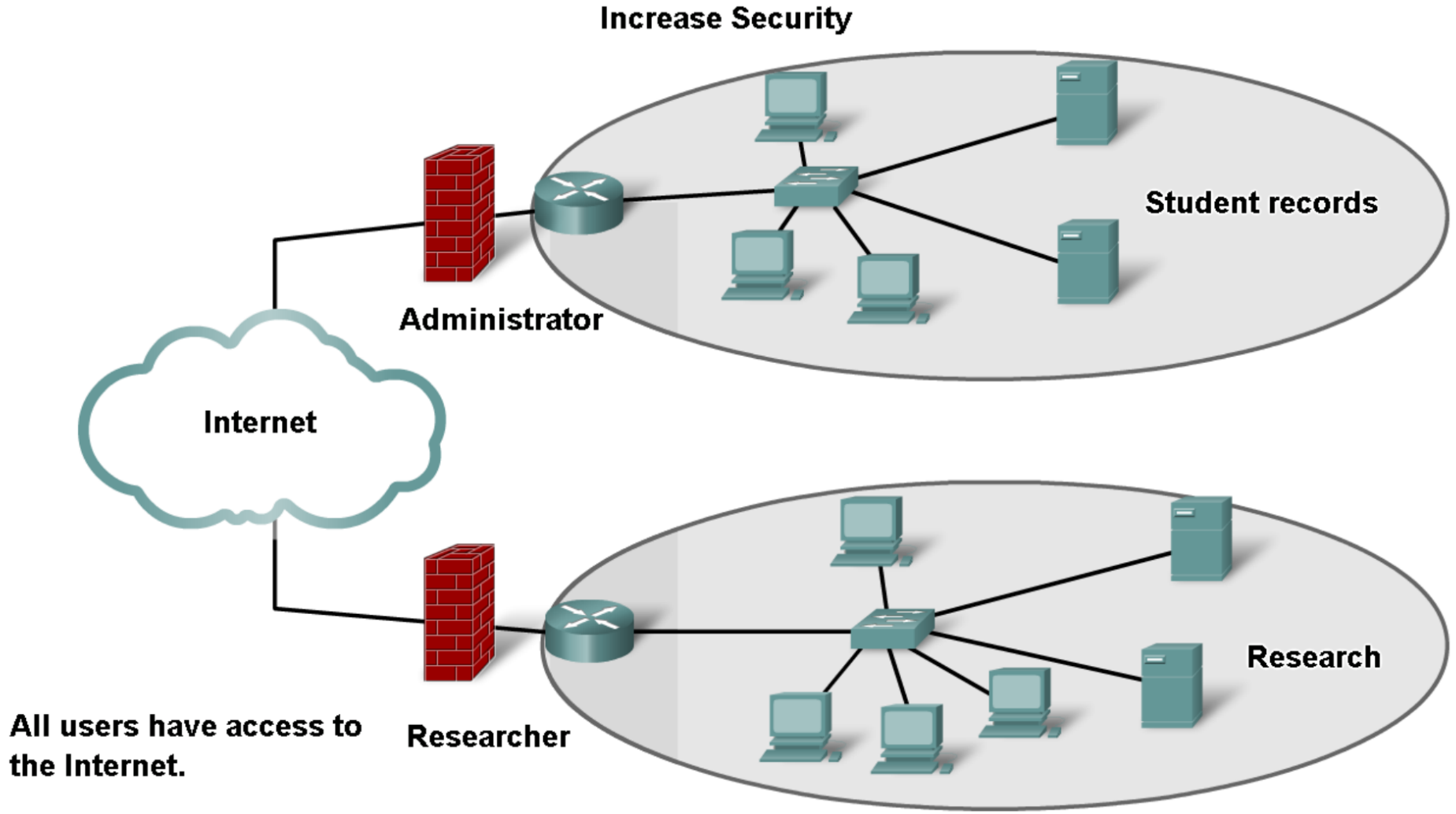


All devices in this network are connected in one broadcast domain when the switch is set to the factory default settings. Since switches forward broadcasts by default, broadcasts are processed by all devices in this network.

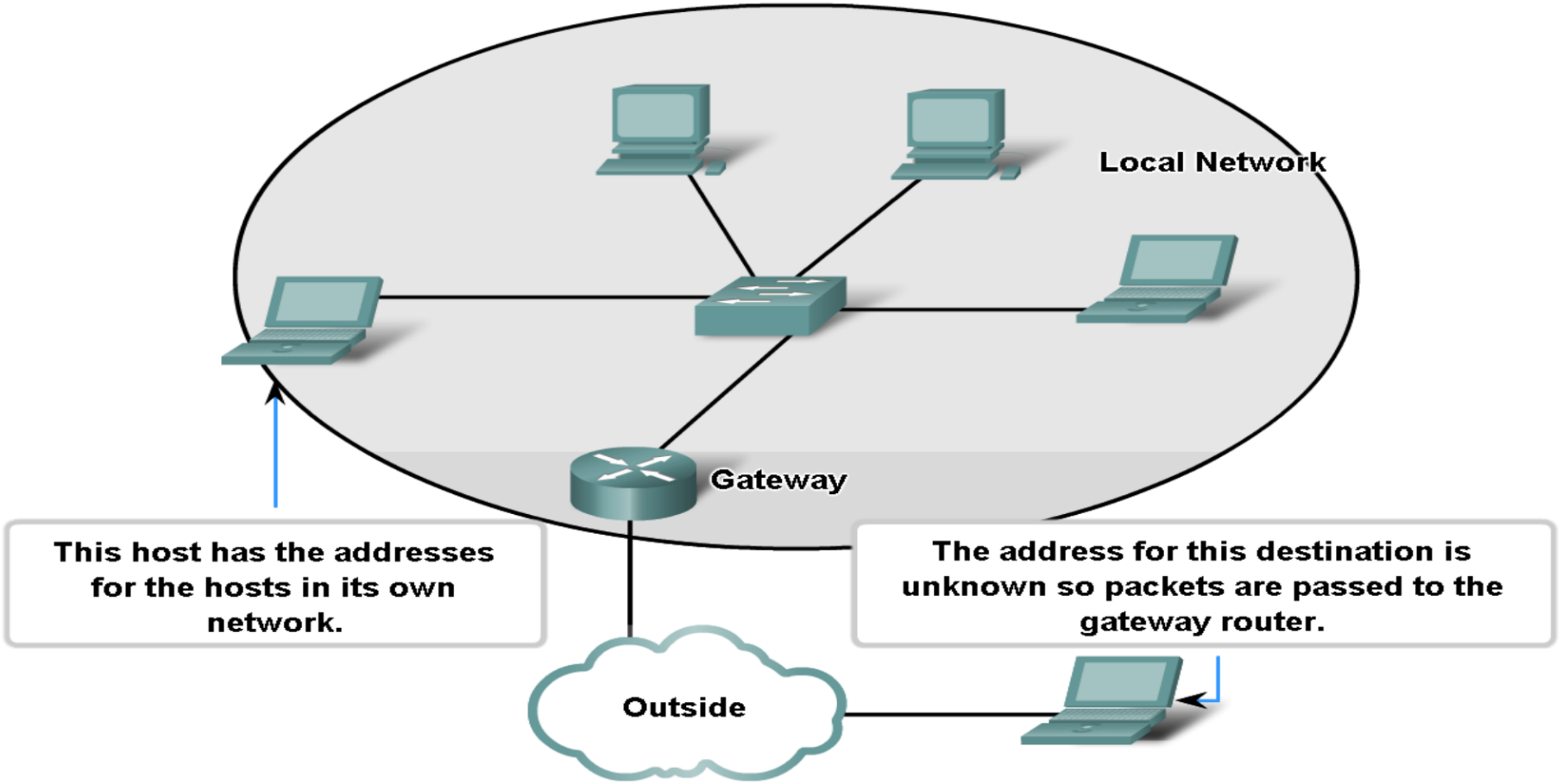


Replacing the middle switch with a router creates 2 IP subnets, hence, 2 distinct broadcast domains. All devices are connected but local broadcasts are contained.

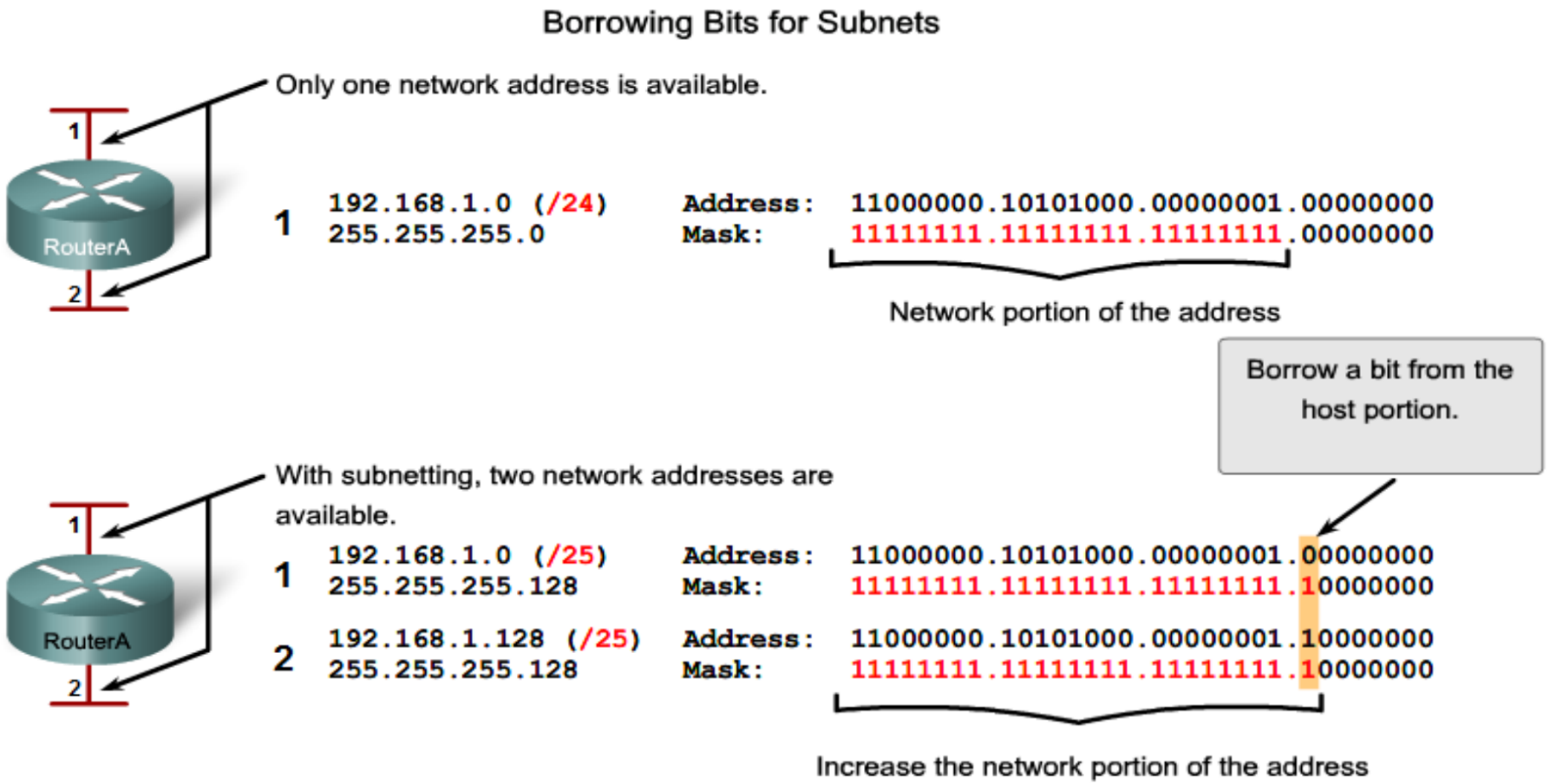
3. IPv4 Subnetting - Increase Network Security



3. IPv4 Subnetting - Increase Network Manageability



3. IPv4 Subnetting - Borrowing Bits



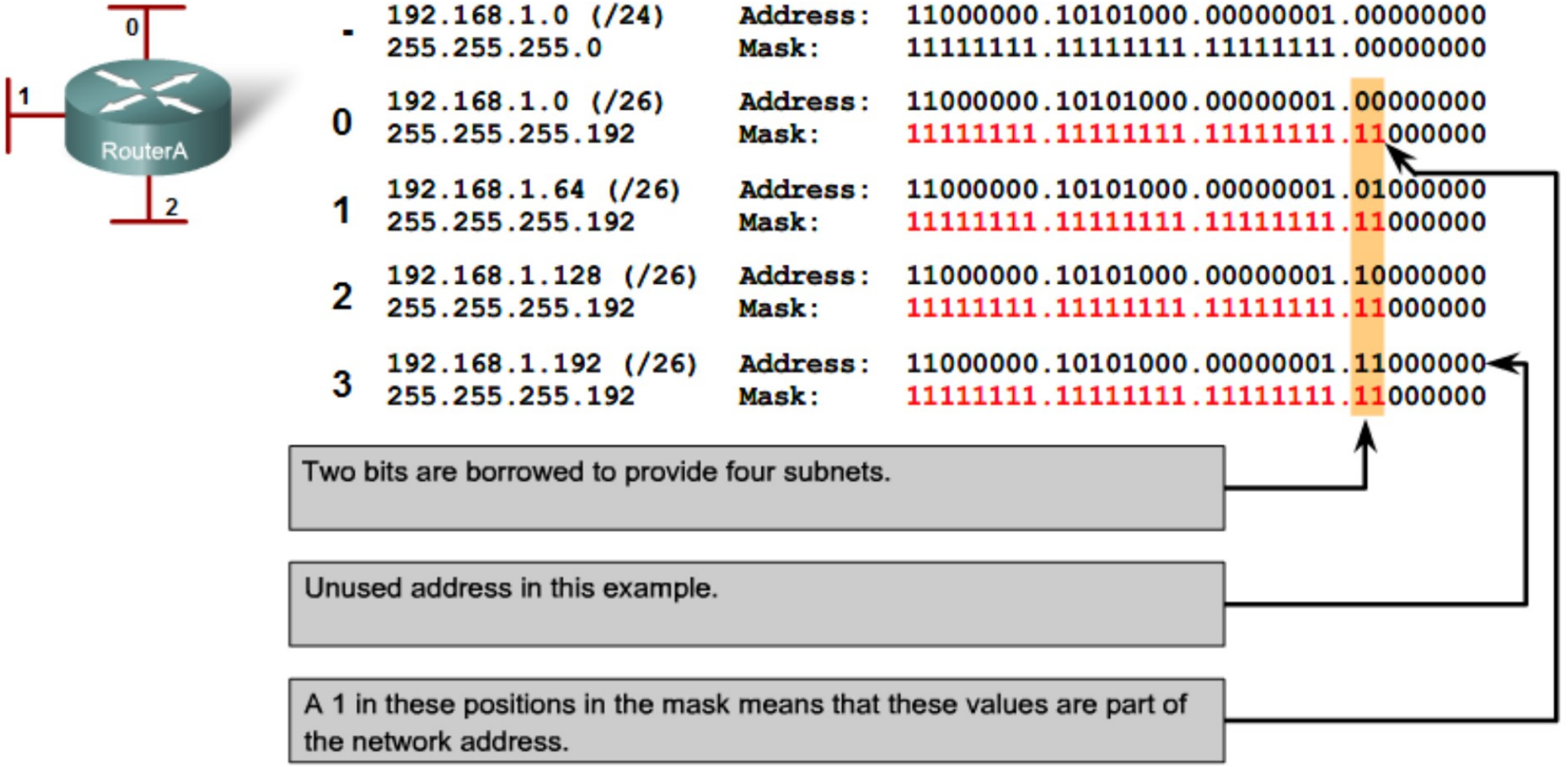
Borrow bits from the host portion → **2^n subnets**, each with **2^h − 2 hosts** (h = remaining host bits).

3. IPv4 Subnetting - Example of 2 Networks

Subnet	Network address	Host range	Broadcast address
0	192.168.1.0/25	192.168.1.1 - 192.168.1.126	192.168.1.127
1	192.168.1.128/25	192.168.1.129 - 192.168.1.254	192.168.1.255

Borrowing **1 bit** → mask **255.255.255.128 (/25)** → **2 subnets** of **126 hosts** each ($2^7 - 2$)

3. IPv4 Subnetting - Borrowing 2 Bits

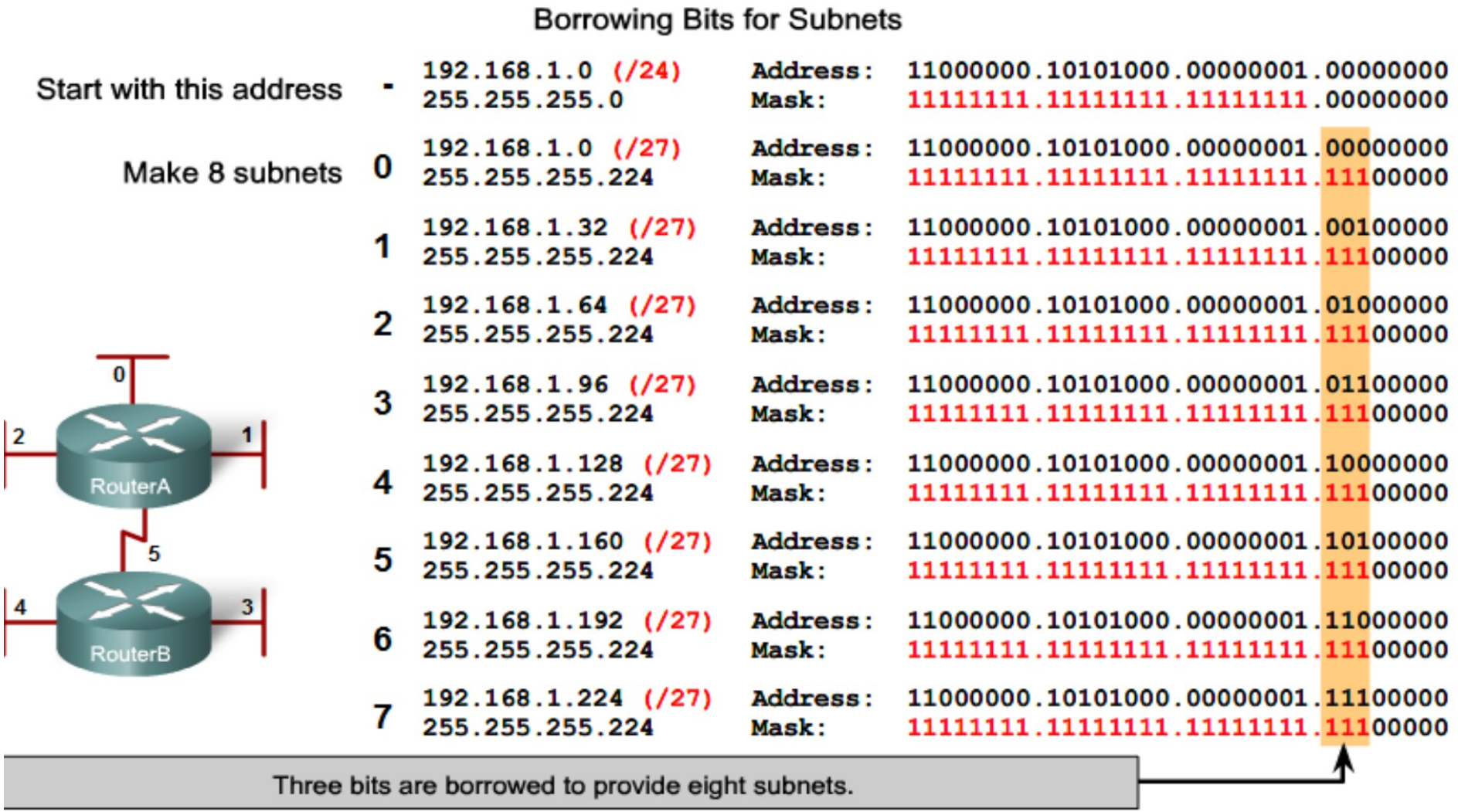


3. IPv4 Subnetting - Example of 4 Networks

Subnet	Network address	Host range	Broadcast address
0	192.168.1.0/26	192.168.1.1 - 192.168.1.62	192.168.1.63
1	192.168.1.64/26	192.168.1.65 - 192.168.1.126	192.168.1.127
2	192.168.1.128/26	192.168.1.129 - 192.168.1.190	192.168.1.191
3	192.168.1.192/26	192.168.1.193 - 192.168.1.254	192.168.1.255

Borrowing **2 bits** → mask **255.255.255.192 (/26)** → **4 subnets** of **62 hosts** each ($2^6 - 2$); block size = 64

3. IPv4 Subnetting - Borrowing 3 Bits



3. IPv4 Subnetting - Example of 8 Networks

Subnet	Network address	Host range	Broadcast address
0	192.168.1.0/27	192.168.1.1 - 192.168.1.30	192.168.1.31
1	192.168.1.32/27	192.168.1.33 - 192.168.1.62	192.168.1.63
2	192.168.1.64/27	192.168.1.65 - 192.168.1.94	192.168.1.95
3	192.168.1.96/27	192.168.1.97 - 192.168.1.126	192.168.1.127
4	192.168.1.128/27	192.168.1.129 - 192.168.1.158	192.168.1.159
5	192.168.1.160/27	192.168.1.161 - 192.168.1.190	192.168.1.191
6	192.168.1.192/27	192.168.1.193 - 192.168.1.222	192.168.1.223
7	192.168.1.224/27	192.168.1.225 - 192.168.1.254	192.168.1.255

Borrowing **3 bits** → mask **255.255.255.224 (/27)** → **8 subnets** of **30 hosts** each ($2^5 - 2$); block size = 32

3. IPv4 Subnetting - Example 1

Which IP address should be assigned to PC B?

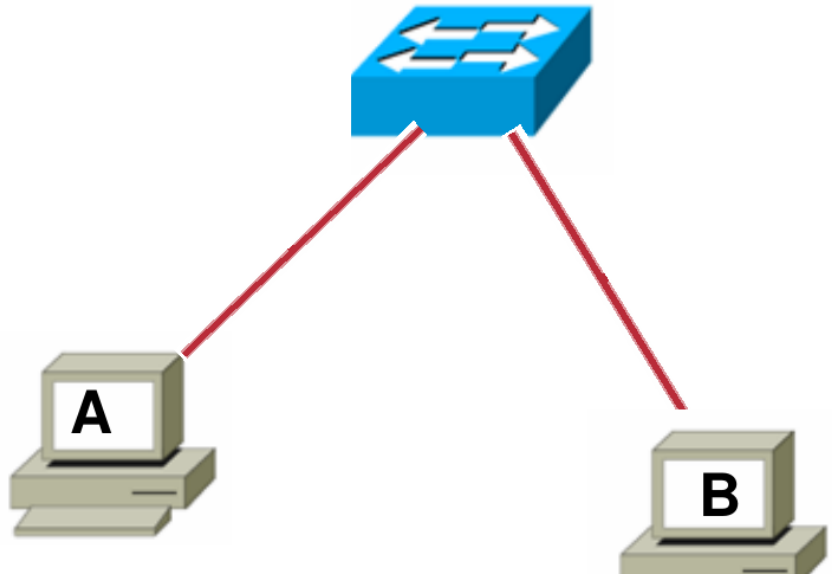
A. 192.168.5.5

B. 192.168.5.32

C. 192.168.5.40

D. 192.168.5.63

E. 192.168.5.75



PC A: 192.168.5.33/27

Answer: C

Solution: /27 → block size 32 → network **192.168.5.32**, broadcast **192.168.5.63**, valid hosts **.33 to .62**. Only **192.168.5.40** is a valid host in the same subnet as PC A.

3. IPv4 Subnetting - Example 2

Given the choices below, which address represents a unicast address?

A. 224.1.5.2

B. FFFF.FFFF.FFFF

C. 192.168.24.59/30

D. 255.255.255.255

E. 172.31.128.255/18

Answer: E

Choice	Why?
A	Class D → multicast
B	Broadcast MAC address (Layer 2), not IPv4
C	/30 → subnet .56 – .59: .59 is the broadcast
D	Limited broadcast address
E	/18 → 172.31.128.0 – 172.31.191.255: .128.255 is a valid host

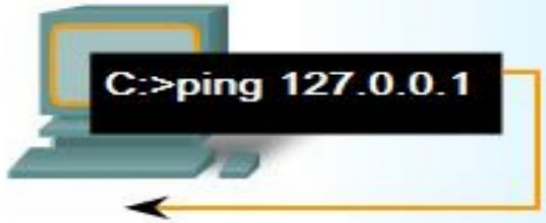
3. IPv4 Subnetting - Test Connectivity

- **Ping** is a utility for testing IP connectivity between hosts. It sends requests for responses to a specified host address.
- Ping uses **ICMP** (Internet Control Message Protocol), a Layer 3 protocol of the TCP/IP suite: it sends an **ICMP Echo Request**.
- If the destination host receives the Echo Request, it responds with an **ICMP Echo Reply**.
- For each packet sent, ping measures the **time required for the reply**.

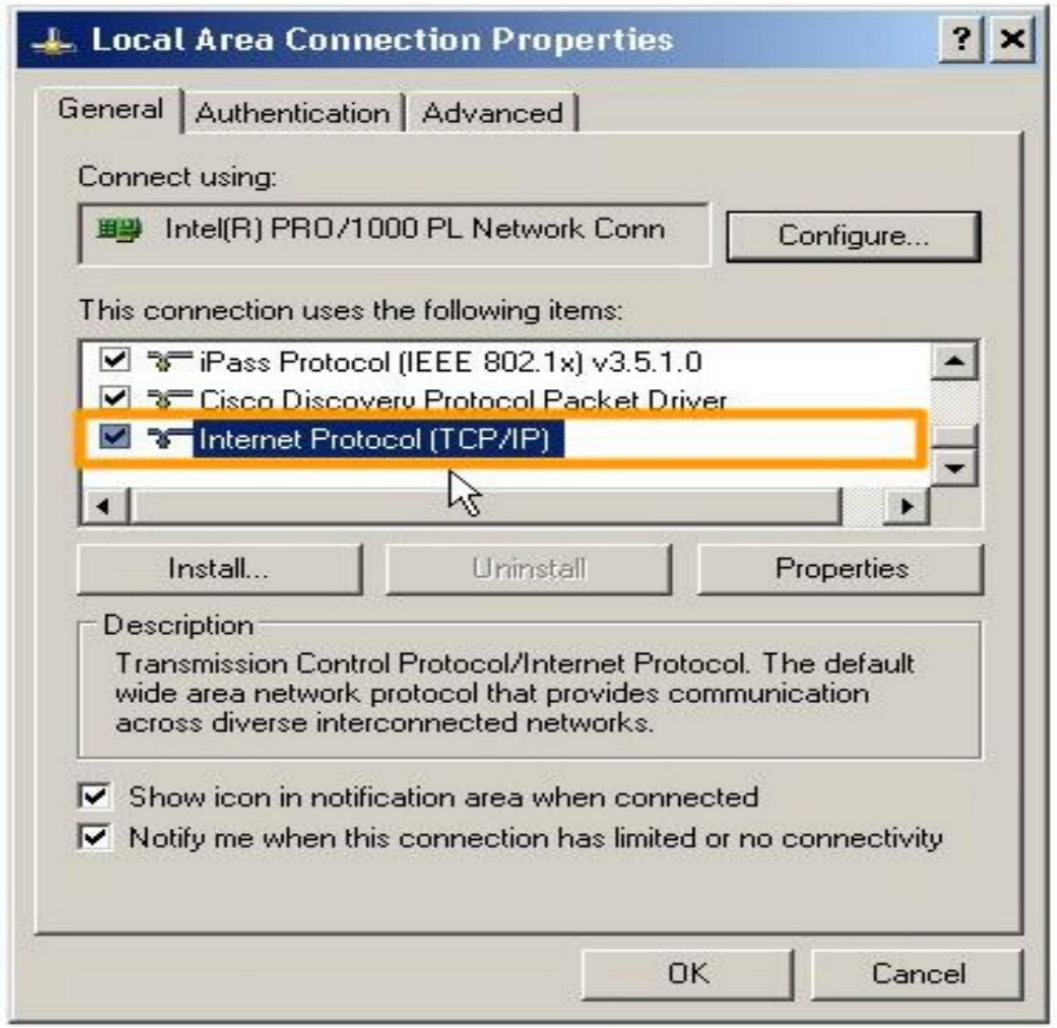
3. IPv4 Subnetting - Testing the Local TCP/IP Stack

Testing Local TCP/IP Stack

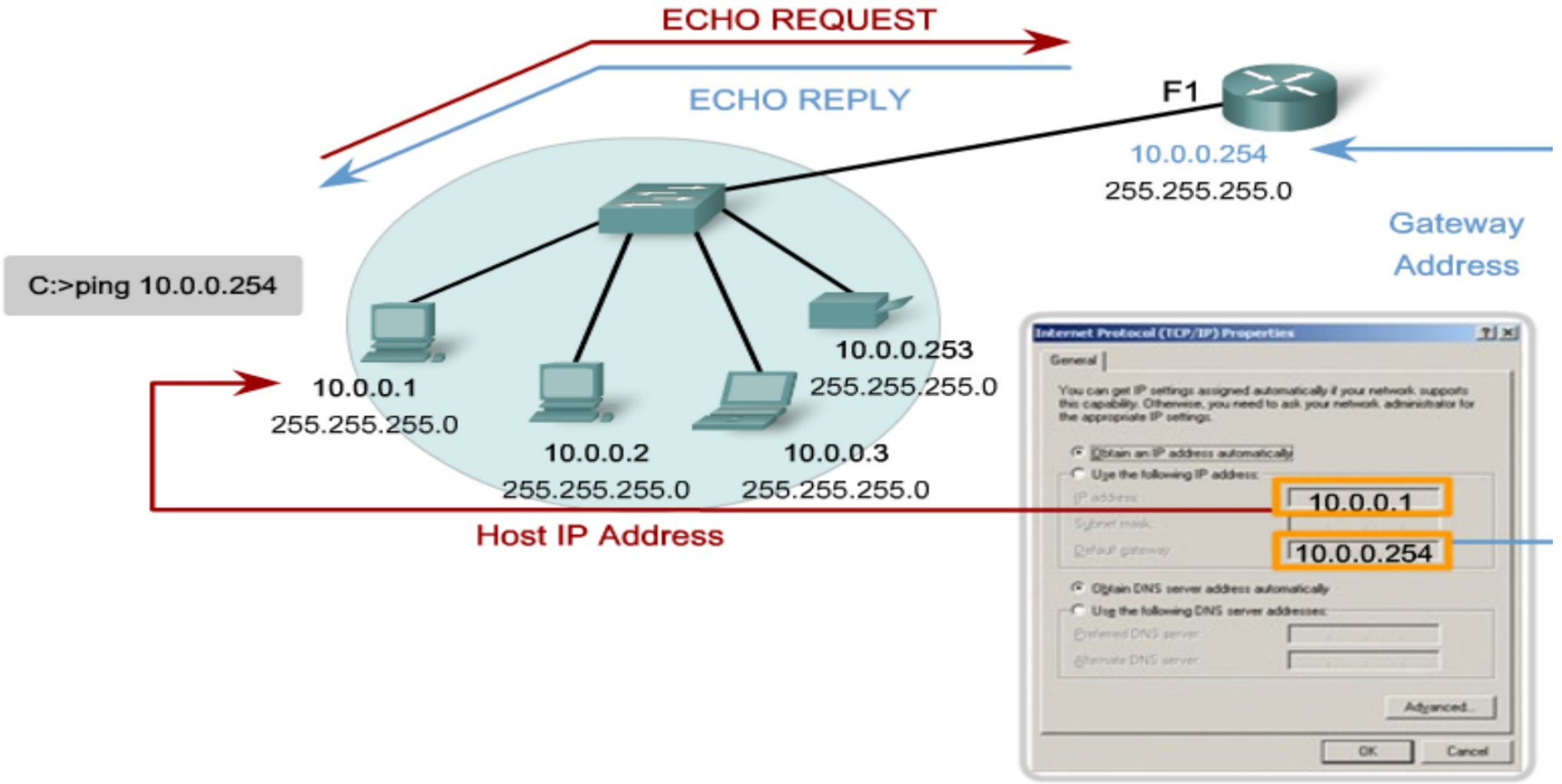
Pinging the local host confirms that TCP/IP is installed and working on the local host.



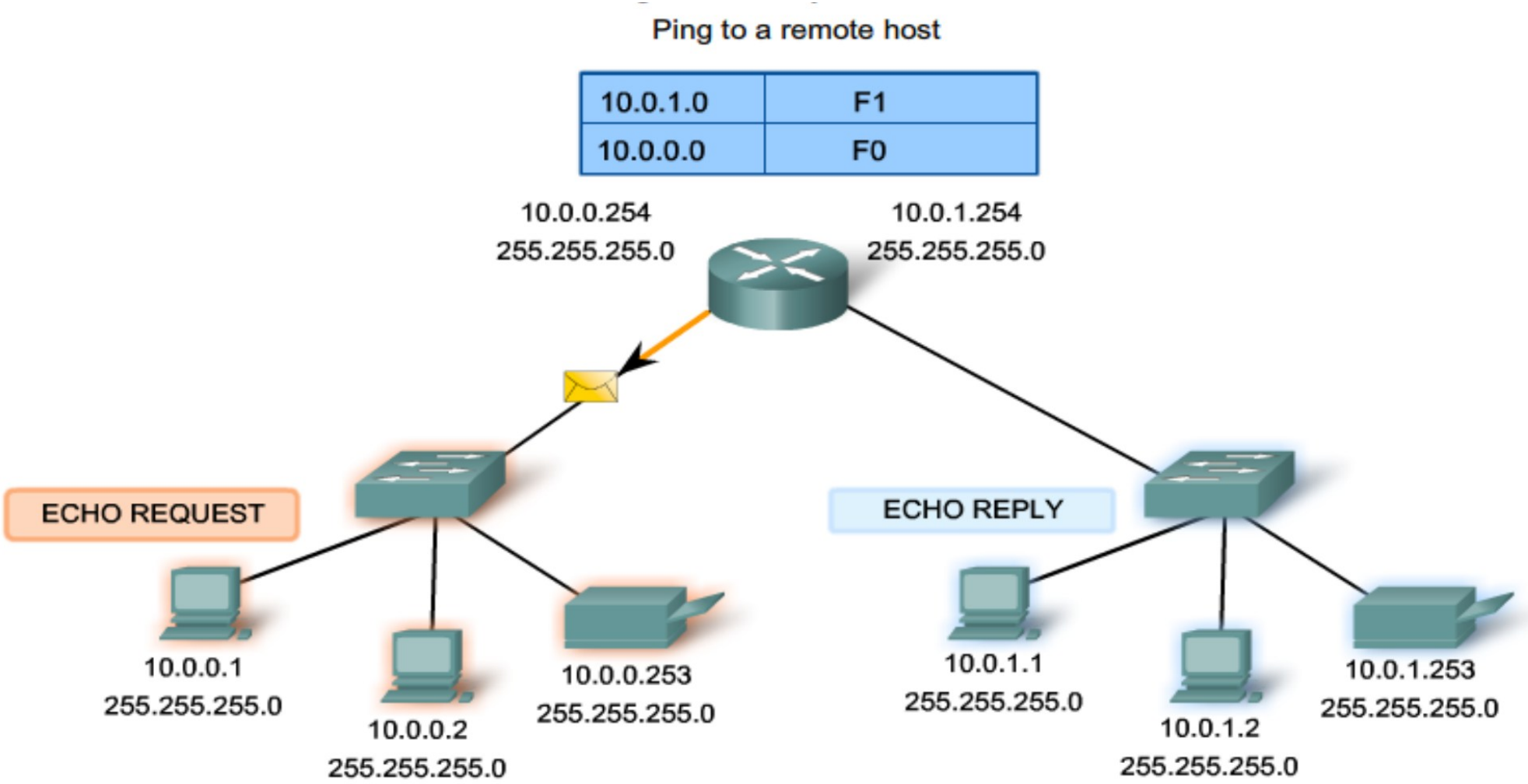
Pinging 127.0.0.1 causes a device to ping itself.



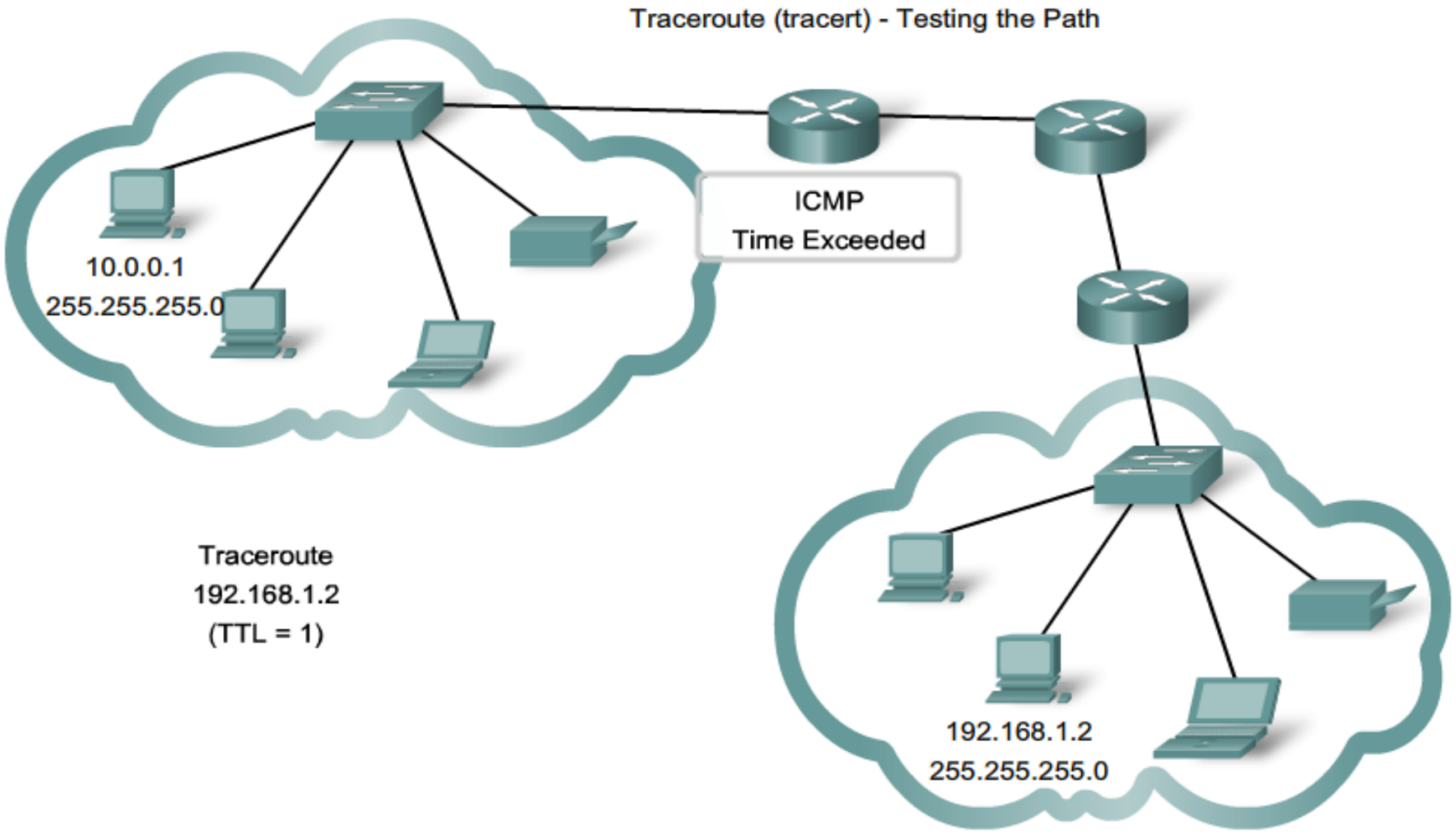
3. IPv4 Subnetting - Testing the Local Network



3. IPv4 Subnetting - Testing a Remote LAN



3. IPv4 Subnetting - Traceroute (tracert)



Each router that decrements the **TTL to 0** returns an **ICMP Time Exceeded** message → the path is discovered hop by hop.

4. VLSM and CIDR - Objectives

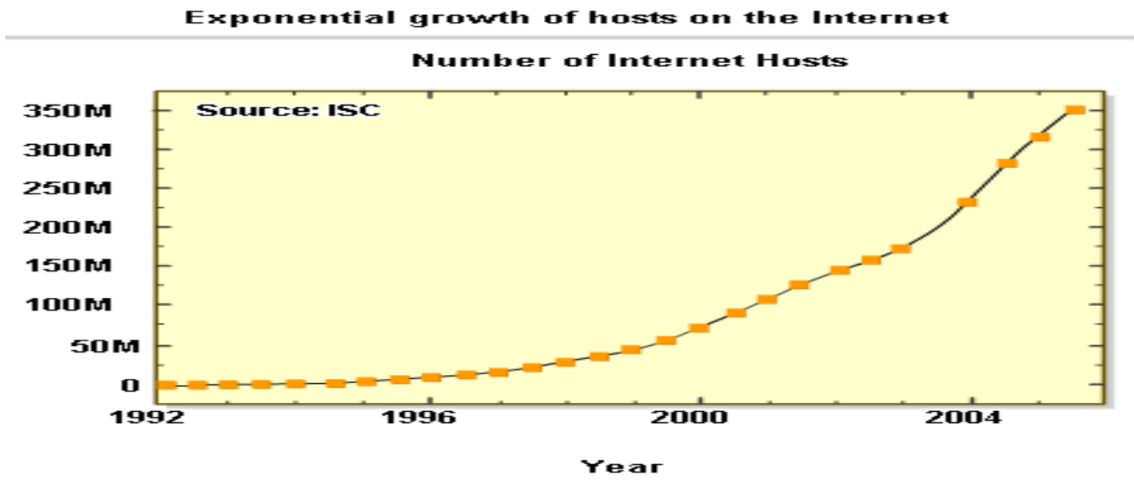
- Compare and contrast **classful** and **classless** IP addressing.
- Review **VLSM** and explain the benefits of classless IP addressing.
- Describe the role of the **Classless Inter-Domain Routing (CIDR)** standard in making efficient use of scarce IPv4 addresses.

4. VLSM and CIDR - Introduction

- Prior to 1981, IP addresses used only the **first 8 bits** to specify the network portion of the address.
- In 1981, **RFC 791** modified the IPv4 32-bit address to allow for **three different classes**.
- The IP address space was depleting rapidly: the Internet Engineering Task Force (IETF) introduced **Classless Inter-Domain Routing (CIDR)**.
  - CIDR uses **Variable Length Subnet Masking (VLSM)** to help conserve address space.
  - **VLSM is simply subnetting a subnet.**

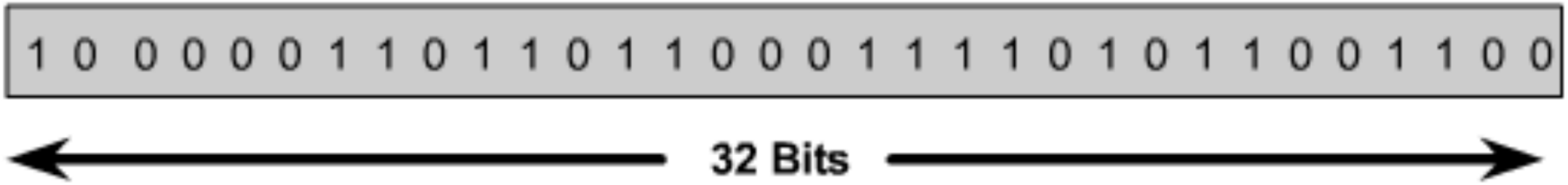
4. VLSM and CIDR - Classful and Classless Addressing

- As of January 2007, there were **over 433 million hosts** on the Internet.
- Initiatives to conserve IPv4 address space include:
  - **VLSM & CIDR** notation (1993, RFC 1519)
  - **Network Address Translation** (1994, RFC 1631)
  - **Private Addressing** (1996, RFC 1918)



4. VLSM and CIDR - The High Order Bits

The **high order bits** are the **leftmost bits** in a 32-bit address.



Binary : 11000000.10101000.000000001.00001000 and 11000000.10101000.00000001.00001001

Decimal : 192.168.1.8 and 192.168.1.9

4. VLSM and CIDR - Classes and High Order Bits

Classes of IP addresses are identified by the decimal number of the **1st octet**:

- **Class A** addresses begin with a **0** bit.
- **Class B** addresses begin with a **1** bit and a **0** bit.
- **Class C** addresses begin with **two 1** bits and a **0** bit.

Class	High order bits	Start	End
Class A	0	0.0.0.0	127.255.255.255
Class B	10	128.0.0.0	191.255.255.255
Class C	110	192.0.0.0	223.255.255.255
Multicast	1110	224.0.0.0	239.255.255.255
Experimental	1111	240.0.0.0	255.255.255.255

4. VLSM and CIDR - Classful Addressing Structure

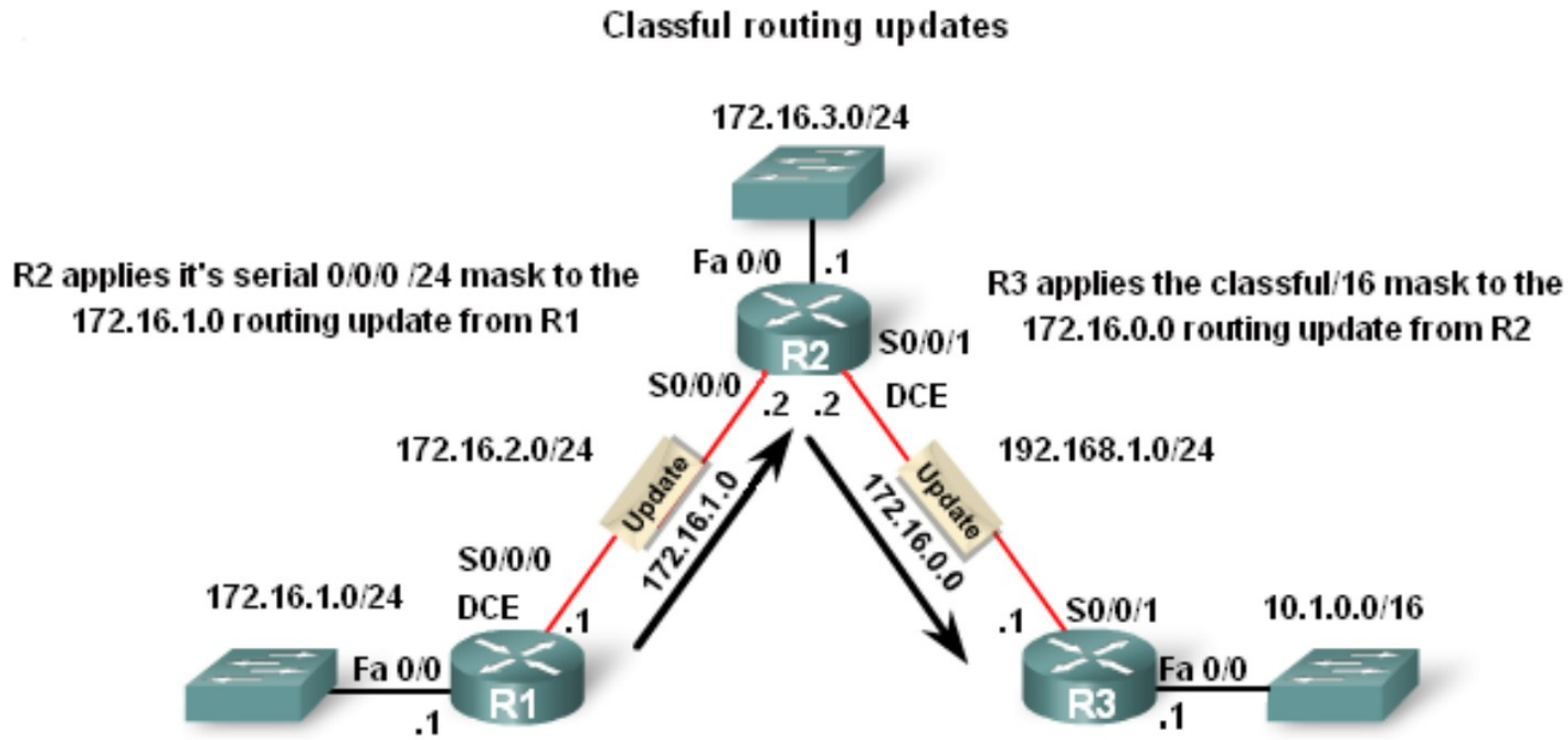
An IP address has 2 parts (RFC 790): the **network** portion on the **left** and the **host** portion on the **right**. The subnet mask determines the network portion.

	1st Octet	2nd Octet	3rd Octet	4th Octet	Subnet Mask
Class A	Network	Host	Host	Host	255.0.0.0 or /8
Class B	Network	Network	Host	Host	255.255.0.0 or /16
Class C	Network	Network	Network	Host	255.255.255.0 or /24

Address class	First octet range	Number of possible networks	Number of hosts per network
Class A	0 to 127	128 (2 are reserved)	16,777,214
Class B	128 to 191	16,384	65,534
Class C	192 to 223	2,097,152	254

4. VLSM and CIDR - Classful Routing Updates

- **Classful routing protocols** (e.g., RIPv1) **do not send subnet masks** in their routing updates.
- The reason: the subnet mask is directly related to the **network address class**.



4. VLSM and CIDR - Classless Inter-Domain Routing

CIDR (RFC 1519)

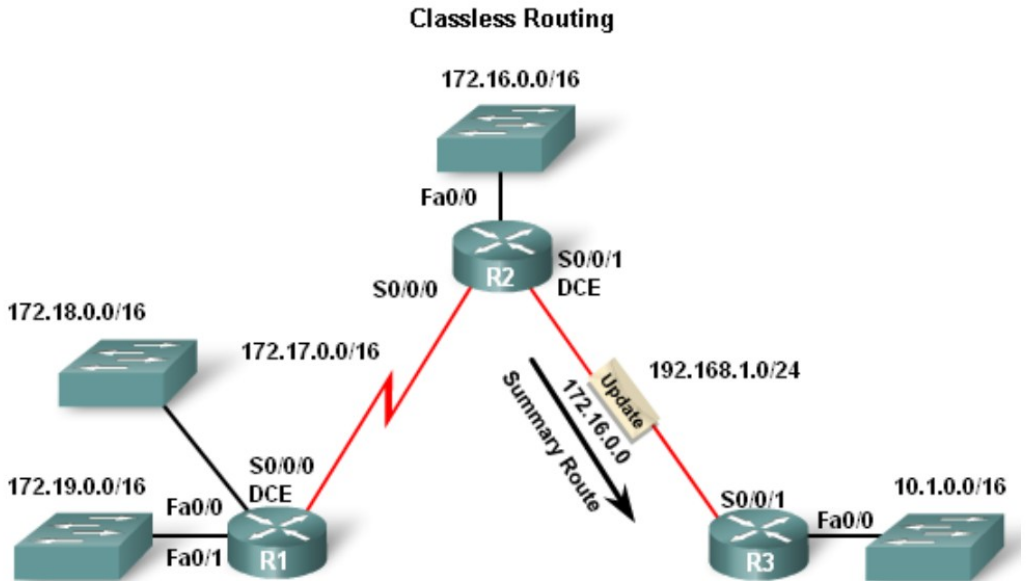
- Advantages of CIDR:
  - More efficient use of IPv4 address space.
  - Route summarization.
- Requires the **subnet mask to be included in routing updates**, because the address class is meaningless.
- Recall the purpose of a subnet mask: to determine the **network** and **host** portions of an IP address.

4. VLSM and CIDR - Classless Routing Protocols

Characteristics of **classless routing protocols**:

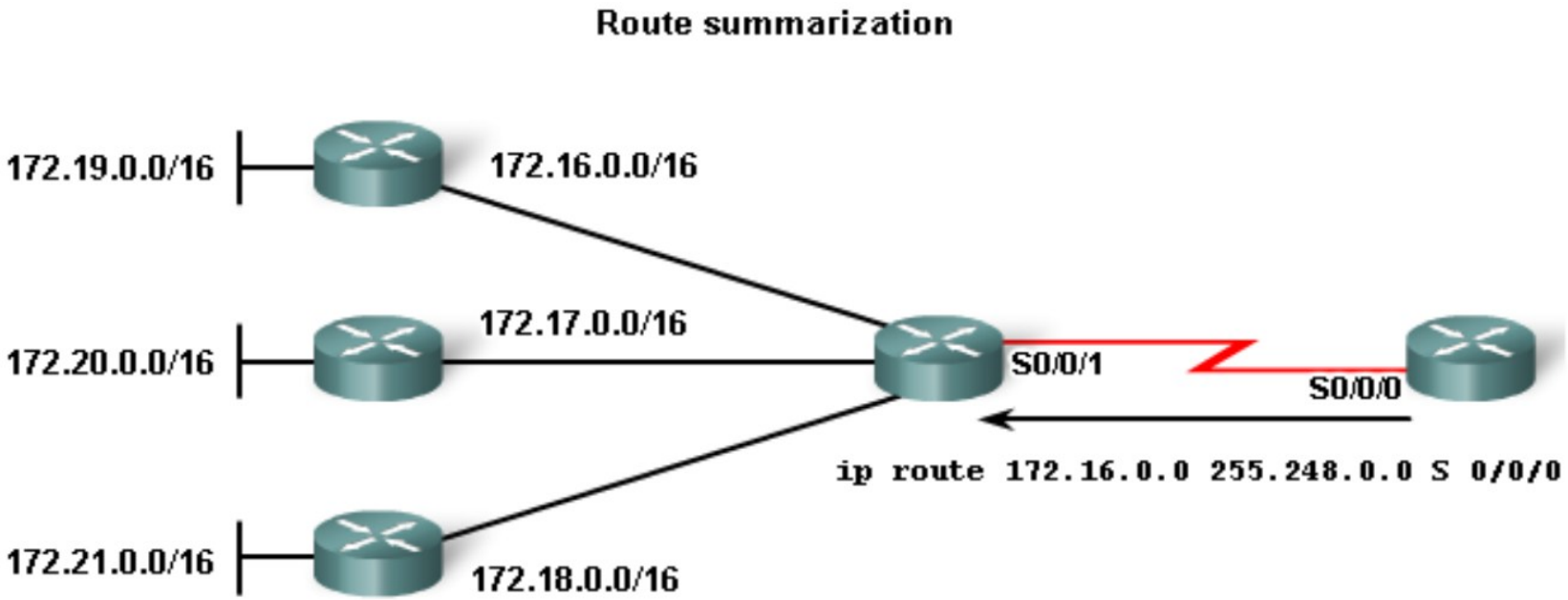
- Routing updates **include the subnet mask**.
- **Supports VLSM.**
- **Supports route summarization.**

Routing protocol	Updates include subnet mask	Supports VLSM	Can send supernet routes
Classful	No	No	No
Classless	Yes	Yes	Yes



4. VLSM and CIDR - Route Summarization

- Routes are summarized with masks that are **less than** the **default classful mask**.
- **Example:** 172.16.0.0 **/13** is the summarized route for the 172.16.0.0 **/16** to 172.23.0.0 **/16** classful networks.



4. VLSM and CIDR - Calculating a Route Summary

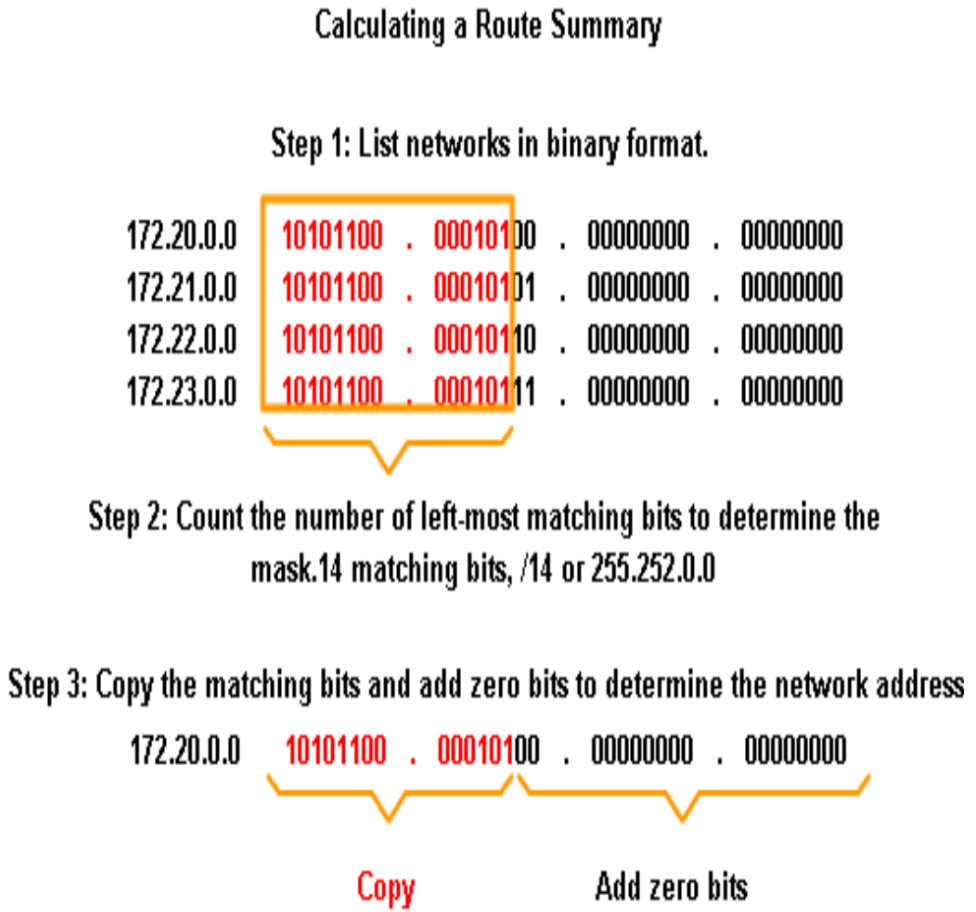
Steps to calculate a route summary:

1. List the networks in **binary** format.

2. Count the number of **left-most matching bits** to determine the summary route's mask.

3. Copy the matching bits and **add zero bits** to determine the summarized network address.

Result: 172.20.0.0/14 (255.252.0.0)



Summary

In this chapter, you learned to:

- Explain the structure of IP addressing and convert between 8-bit binary and decimal numbers.
- Given an IPv4 address, classify it by type and describe how it is used in the network.
- Explain how addresses are assigned to networks by ISPs and within networks by administrators.
- Determine the network portion of the host address and explain the role of the subnet mask in dividing networks.
- Given IPv4 addressing information and design criteria, calculate the appropriate addressing components.
- Use common testing utilities to verify network connectivity and the operational status of the IP stack on a host.
- Explain the benefits of classless addressing, VLSM, and CIDR route summarization.



THE END